

Assignment Week 1

Introduction to Machine Learning-IITKGP

TYPE OF QUESTION: MCQ, MSQ, NUMERIC

Number of Questions: 15

Total Marks: 15x2 = 30

Question 1:

Which of the following is the correct definition of supervised learning?

- A. Learning without any labeled data
- B. Learning from labeled input-output pairs to map inputs to outputs
- C. Learning by maximizing a reward signal from an environment
- D. Learning by finding hidden patterns in unlabeled data

Correct Answer: B. Learning from labeled input-output pairs to map inputs to outputs

Detailed Solution: Supervised learning uses labeled training data i.e. pairs of inputs and corresponding correct outputs to learn a mapping function.

Question 2:

Which metric measures the fraction of actual positives that were correctly identified by the model?

- A. Precision
- B. Recall
- C. Accuracy
- D. F1-Score

Correct Answer: B. Recall

Detailed Solution: Recall (also called Sensitivity or True Positive Rate) = $TP / (TP + FN)$. It measures how many of the actual positive instances the model successfully

retrieved. Precision, by contrast, measures the fraction of predicted positives that were actually positive.

Question 3:

A model trained on a small dataset achieves 99% training accuracy but only 55% test accuracy. Which phenomenon is this an example of?

- A. Underfitting
- B. Overfitting
- C. Optimal fitting
- D. Data leakage

Correct Answer: B. Overfitting

Detailed Solution: Overfitting occurs when a model learns the training data too well, including its noise, and fails to generalize to unseen data. The large gap between training accuracy (99%) and test accuracy (55%) is the classic symptom of high variance / overfitting.

[Numeric Type - Range Answer]

Question 4:

A classifier is tested on 500 samples: 200 are positive, 300 are negative. It predicts 180 positives, of which 140 are truly positive. What is the F1-Score (to 2 decimal places)?

Correct Answer: 0.70 to 0.80

Detailed Solution: TP=140, FP=40, FN=60. Precision = $140/180 \approx 0.778$. Recall = $140/200 = 0.700$. $F1 = 2 \times (0.778 \times 0.700) / (0.778 + 0.700) = 2 \times 0.545 / 1.478 \approx 0.738$.

Question 5:

Suppose a classifier predicts 120 emails as spam, out of which 90 are actually spam emails. The total number of spam emails in the dataset is 150.

What is the precision and recall of the spam classifier?

- A. Precision = 75%, Recall = 60%
- B. Precision = 60%, Recall = 75%
- C. Precision = 75%, Recall = 75%
- D. Precision = 60%, Recall = 60%

Correct Answer: A

Detailed Solution:

$TP = 90$, $FP = (120 - 90) = 30$, $FN = (150 - 90) = 60$

Precision = $TP / (TP + FP) = 90 / (90 + 30) = 75\%$.

Recall = $TP / (TP + FN) = 90 / 150 = 60\%$.

Question 6:

Model A has high bias and low variance. Model B has low bias and high variance. Which statement best characterizes the bias-variance tradeoff?

- A. Model A overfits; Model B underfits
- B. Model A underfits; Model B overfits
- C. Both models have equal generalization error
- D. Bias and variance are independent and can both be minimized simultaneously with no tradeoff

Correct Answer: B. Model A underfits; Model B overfits

Detailed Solution: High bias means the model is too simple and systematically wrong (underfitting). High variance means the model is too sensitive to training data fluctuations (overfitting). The bias-variance tradeoff describes the tension: reducing one typically increases the other for a fixed model capacity.

Question 7:

Which of the following is an example of an unsupervised learning task?

- A. Predicting whether a tumour is malignant or benign
- B. Estimating tomorrow's stock closing price
- C. Segmenting customers into groups based on purchasing behaviour
- D. Recognizing handwritten digits from 0–9

Correct Answer: C. Segmenting customers into groups based on purchasing behaviour

Detailed Solution: Customer segmentation involves finding natural groupings in data without predefined labels — this is clustering, which is unsupervised learning. All other options require labeled training data (supervised learning).

Question 8:

Which of the following scenarios would most benefit from using reinforcement learning instead of supervised learning? Select ALL that apply.

- A. Training a chess-playing agent from scratch
- B. Classifying emails as spam or not spam given 100,000 labelled examples
- C. Training a robot arm to pick and place objects through trial and error
- D. Teaching an autonomous agent to play a new video game with no prior human gameplay data

Correct Answer: A, C, D

Detailed Solution: Reinforcement learning is ideal when: (1) there is no labeled training data but feedback (reward) is available, and (2) the task requires sequential decision making. Chess, robotic manipulation, and novel game playing all fit this profile. Spam classification (B) has labeled data and is better suited to supervised classification.

Question 9:

Which of the following scenarios is best formulated as a Regression task?

- A. Categorizing customer reviews into positive, neutral, or negative sentiments.
- B. Estimating the number of units of a new product that will be sold next month.
- C. Identifying whether an incoming transaction is fraudulent or legitimate.
- D. Organizing a library of unlabelled images into distinct clusters based on visual similarity.

Correct Answer: B

Detailed Solution: Regression tasks involve predicting a continuous numerical output value. Estimating the number of units sold (Option B) outputs a continuous value. Options A and C are classification tasks because they predict discrete categories (sentiments or fraudulent/legitimate). Option D is an unsupervised clustering task, not a regression task.

Question 10:

Which of the following statements about cross-validation are true?

- I. Cross-validation helps estimate model generalization performance.
- II. k-fold cross-validation uses multiple train-test splits.
- III. Cross-validation eliminates overfitting completely.

- A. I only
- B. I and II
- C. II and III
- D. I, II and III

Correct Answer: B

Detailed Solution: Cross-validation helps estimate model generalization performance. Cross-validation evaluates models using multiple train-test splits. However, it does not completely eliminate overfitting.

[Numeric Type - Exact Answer]

Question 11:

Consider a dataset where each instance is defined by 3 independent boolean features. How many unique Boolean functions can theoretically be defined over this 3-dimensional feature space?

Correct Answer: 256

Detailed Solution:

For N features, there are 2^N unique combinations of inputs.

Given $N = 3$, the number of distinct input rows (combinations) in the truth table is $2^3 = 8$ rows.

Each row can output either 0 or 1. Therefore, the total number of unique boolean functions is given by the formula:

$$\text{Total Functions} = 2^{(2^N)}$$

$$\text{Total Functions} = 2^{(2^3)} = 2^8 = 256.$$

Question 12:

Which of the following statements about overfitting is/are true? (Multiple options may be correct)

- A. Overfitting leads to poor generalization on unseen data.
- B. Overfitting usually results in low training error.
- C. Overfitting always improves test accuracy.
- D. Overfitting reduces the complexity of the model.

Correct Answer: A, B

Detailed Solution: Overfitting occurs when a model learns the training data too closely, including noise and irrelevant patterns. Such models generally achieve very low training error but perform poorly on unseen data, resulting in poor generalization. Therefore, statements A and B are correct. Statement C is false because overfitting usually decreases test accuracy. Statement D is false because overfitting is typically associated with highly complex models.

Question 13:

Identify all the features from the list below that represent Categorical variables rather than Continuous variables: [Select ALL that apply]

- A. The blood type of a patient (e.g., A+, O-, B+)
- B. The exact weight of an infant measured in kilograms
- C. The PIN code of a residential house
- D. The annual income of an individual in Rupees

Correct Answer: A, C

Detailed Solution:

Categorical variables represent qualitative groups or distinct labels. Blood type (Option A) is clearly qualitative. PIN codes (Option C) are numerical digits but act as discrete identifiers/labels rather than measurements; mathematical operations like averaging a ZIP code lack meaning, making it a categorical feature. Infant weight (Option B) and annual income (Option D) are continuous quantitative numerical measurements.

Question 14:

Which of the following tasks is/are classification tasks? (Multiple options may be correct)

- A. Estimating the age of a person from a photo
- B. Diagnosing COVID-19 from chest X-ray (positive/negative)
- C. Predicting whether a student will pass or fail
- D. Predicting rainfall in mm

Answer: B, C

Detailed Solution: B and C have discrete class labels. A and D are regression (continuous output).

Question 15:

A data scientist wants to separate a collection of unlabelled web articles into distinct thematic groups (e.g., sports, politics, tech) without using any pre-existing category labels. Which type of machine learning paradigm must be used?

- A. Supervised Learning
- B. Unsupervised Learning
- C. Reinforcement Learning
- D. Semi-Supervised Learning with a target class

Correct Answer: B

Detailed Solution: Unsupervised learning deals with unlabelled data where the goal is to discover hidden patterns, structures, or groupings within the data. Since the web articles do not possess pre-existing category labels and the objective is to group them by similarity, clustering (a subset of unsupervised learning) is the appropriate approach.
